

zkBugs Evaluation Results

Table 1: Tool Performance Summary

Tool	TP	FN	Timeout	Failure	Total	Median Time (s)
circomspect	1	55	0	0	56	0.08
circom_civer	1	13	14	28	56	120.81
picus	0	30	6	20	56	8.20
ecneproject	7	28	4	16	55	27.59
zkfuzz	4	12	22	18	56	20.69
circom_auditor	33	10	4	3	50	1214.32
TOTAL	46	148	50	85	329	21.46

Table 2: Execution Time Statistics

Tool	Total	Min (s)	Max (s)	Median (s)	Mean (s)	NR-Timeout
circomspect	52	0.01	29.68	0.08	1.55	0
circom_civer	14	0.06	552.42	120.81	205.57	14
picus	30	1.24	222.07	8.20	64.20	6
ecneproject	35	11.18	364.59	27.59	86.37	4
zkfuzz	16	0.01	59.47	20.69	19.59	22
circom_auditor	46	207.77	1741.78	1214.32	1109.48	4

Table 3: Bug-Tool Matrix (by Bug ID)

Bug ID	circospect	circom_civer	picus	ecneproject	zkfuzz	circom_auditor
1	FN*	Timeout	FN	FN	Timeout	Undecided*
2	FN	FN	FN	FN	TP*	Failure
3	FN	FN	Failure	FN	TP*	TP
4	FN	Failure	FN	FN	Failure	TP
5	FN	Failure	Failure	TP*	Failure	TP
6	FN*	Failure	FN	FN	Failure	TP
7	FN	TP	Failure	TP*	TP*	TP
8	FN	Failure	Failure	Failure	Failure	TP
9	FN	Timeout	FN	FN	Timeout	FN*
10	FN	Timeout	FN	FN	Timeout	Timeout
11	TP	Timeout	FN	FN	Timeout	TP*
12	FN	FN	Timeout	Failure	Timeout	TP
13	FN	FN	Failure	Failure	Failure	TP*
14	FN	FN	Failure	Failure	Failure	TP
15	FN	Timeout	FN	FN	Timeout	TP*
16	FN*	Failure	Failure	Timeout	Timeout	FN*
17	FN*	Failure	Timeout	Failure	Failure	TP
18	FN*	FN	Failure	Timeout	Failure	TP*
19	FN*	FN	Failure	Unknown	Failure	Undecided*
20	FN*	FN	Failure	Timeout	Failure	FN*
21	FN*	FN	Failure	Failure	Failure	TP*
22	FN	Failure	FN	FN	FN	TP
23	FN	Failure	FN*	Failure	FN*	FN*
24	FN	Failure	FN	FN	FN	FN*
25	FN	Failure	FN	FN	Timeout	Unknown
26	FN	Failure	FN*	Failure	FN*	TP*
27	FN	Failure	FN*	FN*	FN*	Unknown
28	FN	FN	FN*	FN*	Timeout	TP*
29	FN	Failure	FN*	FN*	FN	TP*
30	FN	Timeout	Failure	TP*	Timeout	FN
31	FN	Timeout	FN	FN	TP*	Unknown
32	FN	FN	FN	FN	Timeout	TP
33	FN	Failure	FN	FN	FN	TP
34	FN	Failure	FN	FN	Timeout	TP
35	FN	Timeout	FN*	FN*	Timeout	Undecided
36	FN*	FN	FN	FN	Timeout	TP
37	FN*	FN	Failure	Timeout	Failure	TP
38	FN	Timeout	Failure	FN	Timeout	TP*
39	FN	Failure	FN	Failure	FN	FN*

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Bug ID	circomspect	circom_civer	picus	ecneproject	zkfuzz	circom_auditor
40	FN	Failure	FN	FN	FN	TP*
41	FN	Failure	FN	FN	FN	TP
42	FN*	Timeout	Failure	Failure	Failure	TP
43	FN	Failure	FN	Failure	FN	FN*
44	FN	Failure	FN	FN	FN	Timeout
45	FN	Timeout	FN	FN	Timeout	TP
46	FN	Timeout	FN	FN	Timeout	TP*
47	FN	Timeout	FN	FN	Timeout	TP*
48	FN	Failure	Timeout	Failure	Failure	FN*
49	FN	Failure	Timeout	Failure	Failure	FN*
50	FN*	Failure	Failure	TP*	Timeout	Failure
51	FN	Failure	Failure	TP*	Timeout	Timeout
52	FN	Failure	Timeout	Failure	Failure	TP*
53	FN	Failure	Failure	TP*	Timeout	Timeout
54	FN*	Failure	Failure	TP*	Timeout	Failure
55	FN	Timeout	Timeout	Failure	Failure	TP*
56	FN	Failure	Failure	Failure	Failure	TP*

* = *manually analyzed*

Table 4: Bug ID to Name Mapping

ID	Bug Name
1	daira_hopwood_darkforest_v0_3_missing_bit_length_check
2	koukyosyumei_checkdepth_comparator_overflow
3	koukyosyumei_clamp_comparator_overflow
4	koukyosyumei_emitifinrange_lessthan_overflow
5	koukyosyumei_extractstringfrompoint_shiftedfirstbyte_unconstrained
6	koukyosyumei_spend_missing_range_check
7	koukyosyumei_subtract_unconstrained_multiplier
8	koukyosyumei_unconstrained_base64_decoded_len
9	koukyosyumei_under_constrained_date_encoder
10	leastauthority_Circuit_Does_Not_Check_the_ERC_20_Sum_Correctly_
11	leastauthority_previously_correct_ownership_proof_disabled_via_code_changes
12	matterlabs_email_spoofing_via_manipulated_from_header
13	matterlabs_emailauth_fails_with_overlapping_invitation_code_regex
14	matterlabs_fromaddrregex_allows_email_address_spoofing
15	matterlabs_underconstrained_fpmul_circuit
16	openzeppelin_mismatched_base64url_decoding_may_break_completeness
17	openzeppelin_non_determinism_of_some_inputs
18	trailofbits_incorrect_handling_of_point_doubling_can_allow_signature_forgery
19	trailofbits_prover_can_lock_user_funds_by_including_ill-formed_bigints_in_public_key_commitment
20	trailofbits_prover_can_lock_user_funds_by_supplying_non-reduced_Y_values_to_G1BigIntToSignFlag
21	veridise_arrayxor_is_under_constrained
22	veridise_babyjubjub_suborder_constraints_not_applied_correctly
23	veridise_blacklist_states_not_representable_in_field
24	veridise_bypassing_internal_transfer_limits_via_swaps
25	veridise_creditnoteproofs_can_be_stolen
26	veridise_data_escrow_encrypted_message_wrong_input
27	veridise_fortxreward_abstracted_away_in_reward_calc
28	veridise_incorrect_initialization_in_membership_circuits
29	veridise_kyt_signature_verification_fails_nonzero_hash
30	veridise_missing_range_checks_in_bigmod
31	veridise_missing_range_checks_on_comparison_circuits
32	veridise_no_zero_value_validation
33	veridise_nullifier_verification_can_be_disabled
34	veridise_ownershipproofs_could_identify_issuers
35	veridise_private_information_leakage
36	veridise_spammers_may_slash_themselves
37	veridise_template_CoreVerifyPubkeyG1_does_not_perform_input_validation_simplified
38	veridise_underconstrained_circuit_allows_invalid_comparison
39	veridise_unsafe_num2bits_254_blacklist_leaf

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ID	Bug Name
40	veridise_zaccount_renewal_kyc_expiry_not_validated
41	veridise_zaccount_renewal_multiple_nullifiers_same_utxo
42	veridise_zero_padding_for_sha256_in_ExpandMessageXMD_is_vulnerable_to_an_overflow
43	veridise_zone_id_inclusion_prover_bypass
44	veridise_zone_limits_bypass_zswapv1
45	yacademy_Knowledge_of_any_member_signature_allow_to_generate_proof_of_membership
46	yacademy_input_signal_s_is_not_constrained_in_eff_ecdsa_circom
47	yacademy_under_constrained_circuits_compromising_the_soundness_of_the_system
48	zksecurity_an_attacker_can_craft_a_fake_non_inclusion_proof_for_a_given_key_due_to_an_aliasing_bug_in_the_smt_verifier
49	zksecurity_big_integer_zero_check_is_not_sound
50	zksecurity_exclusion_check_of_forbidden_countries_is_unsound_and_incomplete_due_to_incorrect_indexing
51	zksecurity_forbidden_country_check_bypass_via_packed_byte_overflow
52	zksecurity_missing_boolean_constraints_in_the_merkle_tree_path_leads_to_an_attacker_being_able_to_craft_a_fake_merkle_proof_for_an_arbitrary_L
53	zksecurity_missing_byte_range_checks_allows_packed_data_pollution
54	zksecurity_second_pre_image_attacks_on_packbytesandposeidon_may_be_used_to_register_arbitrary_passports_and_dsc_certificates
55	zksecurity_sha256_templates_return_zero_on_arbitrary_inputs
56	zksecurity_the_registration_and_disclosure_circuits_lack_range_checks_for_the_input_indices

Table 5: Execution Times per Bug (by Bug ID, in seconds)

Bug ID	circomspect	circom_civer	picus	ecneproject	zkfuzz	circom_auditor
1	0.03	500.81	1.82	14.23	—	882.53
2	0.03	0.27	1.48	19.37	0.05	—
3	0.02	0.20	—	13.05	0.02	494.14
4	0.02	—	1.68	14.31	—	579.94
5	—	—	—	14.69	—	540.69
6	0.03	—	1.45	14.15	—	805.10
7	—	0.06	—	11.18	0.01	207.77
8	0.01	—	—	—	—	1306.21
9	0.01	—	7.14	43.18	—	844.68
10	0.08	—	6.11	27.59	—	—
11	0.12	—	2.45	20.24	—	411.27
12	—	87.14	—	206.83	—	1292.63
13	0.09	458.20	—	202.43	88.93	1543.59
14	—	88.53	—	234.77	119.11	1082.76
15	0.01	—	36.84	160.82	—	918.52
16	0.02	—	—	—	—	1741.78
17	0.02	—	—	303.10	63.00	1212.95
18	0.45	153.08	—	—	316.49	1049.50
19	0.30	538.01	—	—	152.00	1322.24
20	0.39	552.42	—	—	510.74	1359.59
21	0.28	382.08	—	274.24	582.87	1617.11
22	0.16	—	147.87	299.79	20.75	1270.41
23	0.13	—	148.66	152.89	24.19	1495.99
24	0.18	—	118.93	195.36	20.63	1522.81
25	0.02	—	9.25	55.83	—	—
26	0.13	—	143.64	186.85	24.46	1218.27
27	0.18	—	202.28	345.87	22.04	—
28	0.07	346.74	6.17	32.20	—	543.80
29	0.14	—	222.07	333.07	17.64	1722.94
30	0.03	—	—	18.47	—	1106.37
31	0.01	504.04	1.24	13.43	0.36	—
32	0.07	5.24	2.11	15.45	—	420.01
33	0.14	—	218.33	364.59	21.46	1566.33
34	0.09	—	1.58	18.88	—	394.66
35	0.08	—	11.03	40.65	—	846.69
36	0.09	3.20	1.69	14.27	—	226.40
37	0.50	262.77	—	—	166.88	1019.40
38	0.02	536.84	—	13.54	—	636.02
39	0.13	—	182.36	206.56	21.98	1597.62

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Bug ID	circomspect	circom_civer	picus	ecneproject	zkfuzz	circom_auditor
40	0.12	—	40.45	90.91	59.47	1621.88
41	0.20	—	24.45	68.21	41.47	1636.51
42	0.30	—	—	543.44	139.74	1613.73
43	0.15	—	163.60	216.89	20.53	1409.11
44	0.14	—	211.95	363.60	18.42	—
45	0.02	—	3.25	23.91	—	1026.36
46	0.01	—	3.09	23.27	—	1021.03
47	0.01	—	3.12	23.28	—	1215.69
48	29.68	—	—	476.77	121.23	1525.38
49	16.58	—	—	280.75	106.18	1381.19
50	0.04	—	—	100.38	—	—
51	0.03	—	—	62.04	—	—
52	14.71	—	—	428.08	173.59	1423.96
53	0.02	—	—	60.00	—	—
54	0.04	—	—	92.98	—	—
55	0.03	—	—	211.91	80.72	727.23
56	14.28	—	—	284.53	180.84	1633.07